



POLITECNICO
MILANO 1863

ASSIGNMENT 1

STRUCTURAL DYNAMICS AND AEROELASTICITY

Numerical Results

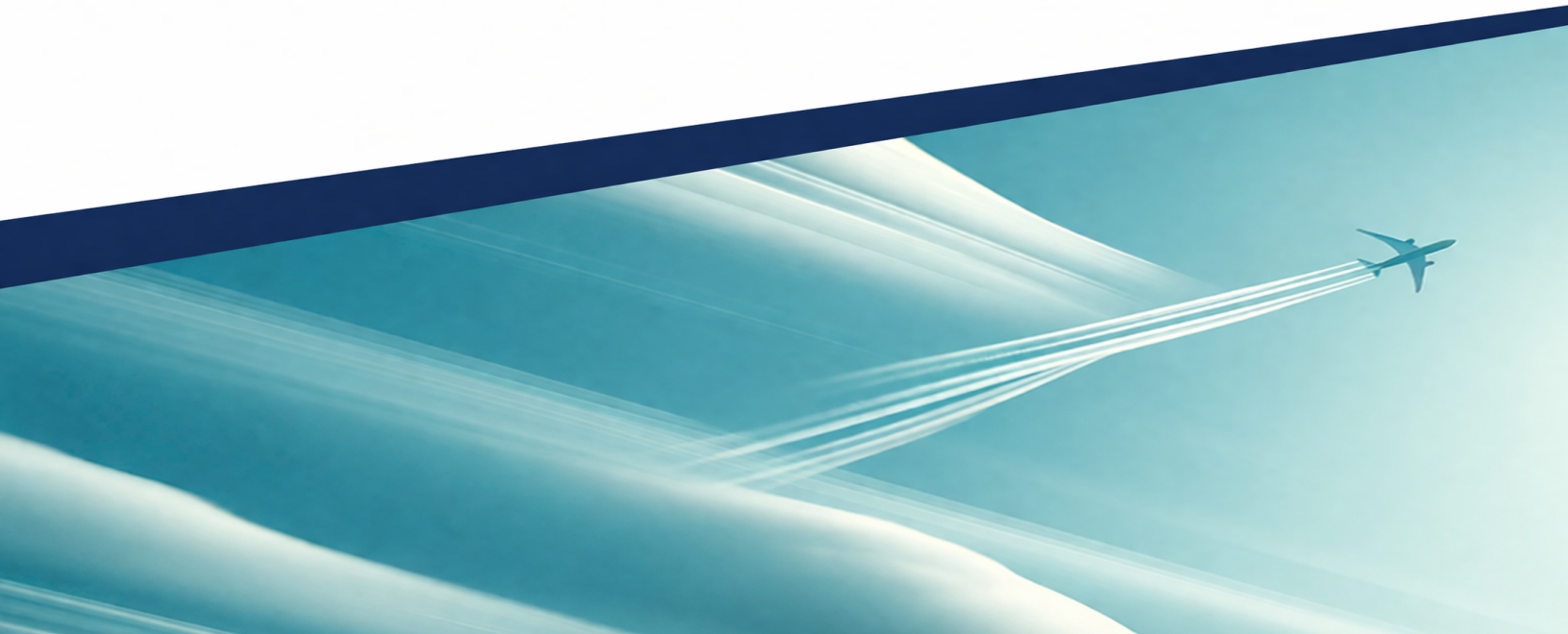
POLITECNICO DI MILANO
Laurea Magistrale – Aeronautical Engineering

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Influence of strut position on divergence

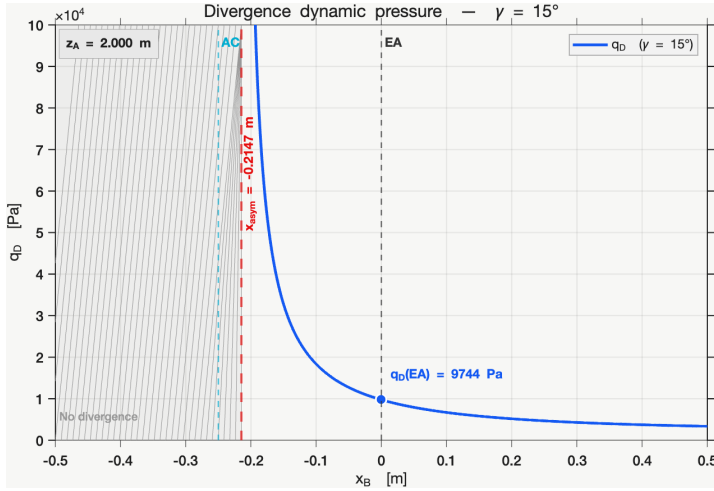


FIGURE 1 – q_D vs. x_B for $\gamma = 15^\circ$.

q_D is highly sensitive to x_B . A vertical asymptote at $x_B \simeq -0.2147$ m separates two regimes :

- **Right** : divergence at positive q_D , dropping rapidly toward the elastic axis ($q_D \approx 9.7 \times 10^3$ Pa).
- **Left** : flexural-torsional stabilization suppresses static divergence entirely.

Limitations. No-divergence region holds within the linear model only – flutter and control reversal remain unconstrained. Forward strut placement may incur buckling/-mass penalties and is sensitive to geometric tolerances near the asymptotic boundary.

Influence of γ on the control effectiveness

x_B fixed; $E_C = L/L_R$ with $E_C = 1$ rigid limit, $E_C > 1$ amplification, $E_C < 0$ reversal. All curves start at $E_C = 1$. Three regimes :

- **Low** γ ($10^\circ, 12.5^\circ$) : $E_C > 1$ – elastic twist amplifies aileron effect – then blocked by divergence; displayed value is actually q_D , not q_C .
- **Intermediate** (15°) : monotonic decrease, genuine reversal at $q_C = 29\,702$ Pa.
- **High** γ ($17.5^\circ, 20^\circ$) : detrimental coupling from low q ; $q_C = 24\,685$ and $21\,427$ Pa – reversal pressure decreases with γ .

$E_C \sim [\det \mathbf{K}_{TOT}]^{-1}$; negative branch past divergence is a mathematical artefact. γ controls sign and magnitude of aeroelastic feedback : low $\gamma \rightarrow$ favorable amplification (divergence governs); high $\gamma \rightarrow$ early control reversal.

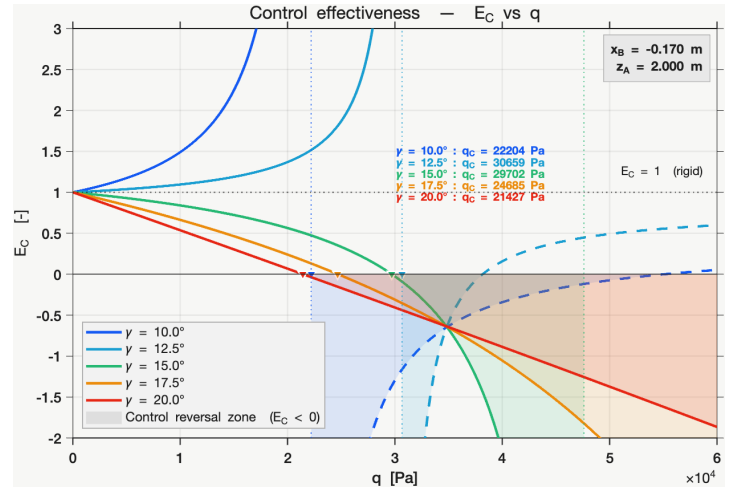
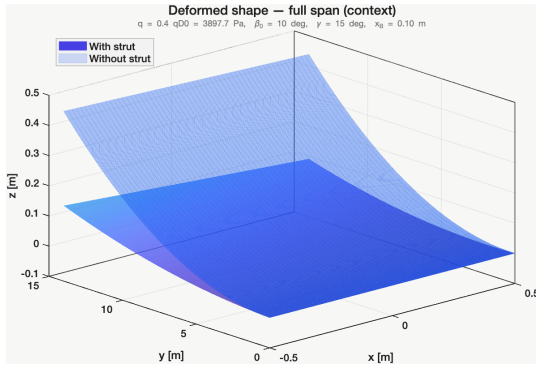


FIGURE 2 – E_C vs. q for several γ , x_B fixed.

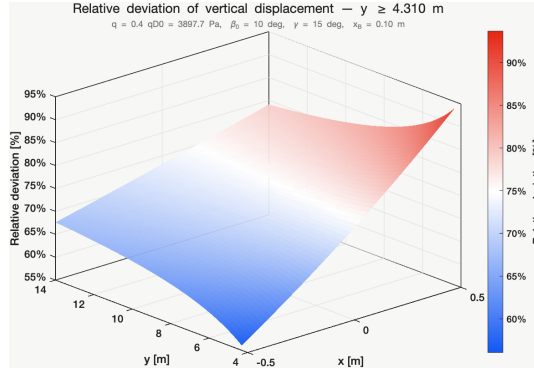
Comparison of the deformed shape with and without the strut

Config : $x_B = 0.10$ m, $\beta_0 = 10^\circ$, $q = 0.4 q_{D,0}$ (subcritical – differences attributable to strut stiffening, not instability proximity). δ_z interpreted only for $y \geq y_{\min} = 4.31$ m ($\approx 30.8\%$ b), defined as the first station where the absolute gap at $x = c/4$ reaches 10% of its tip value, to avoid the root singularity ($z_{ns} \rightarrow 0$).

Both configurations share the same bending-torsion pattern (zero at root, increasing toward tip); the strut reduces amplitude without altering shape :



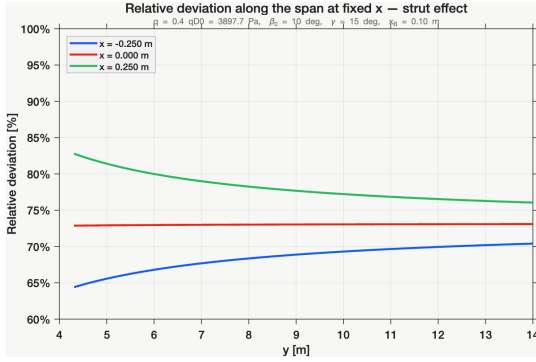
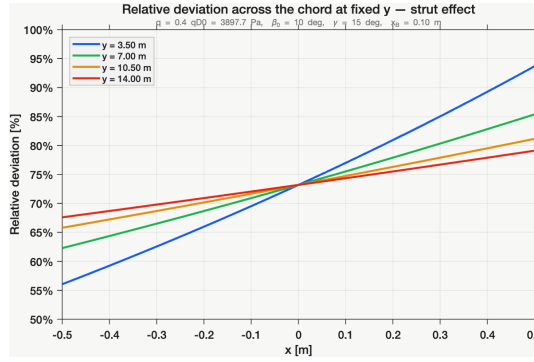
3D shape, with vs. without strut.


 δ_z map, $y \geq y_{\min}$.

1) $\delta_z \in \sim [55\%, 95\%]$ over valid span – strut wing is $\frac{1}{4}-\frac{1}{3}$ as deformed as no-strut reference.

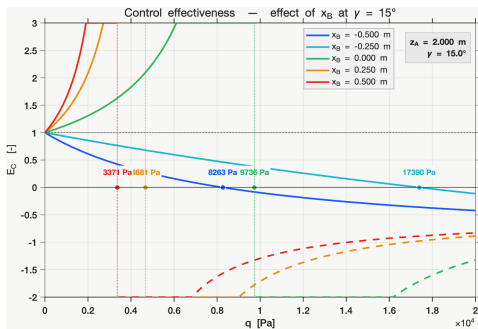
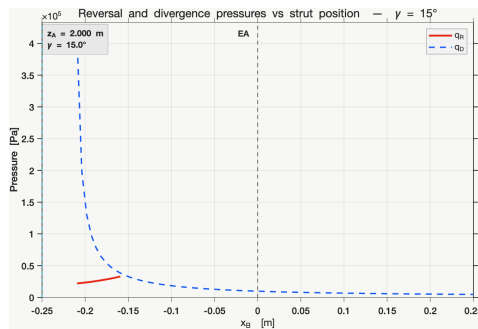
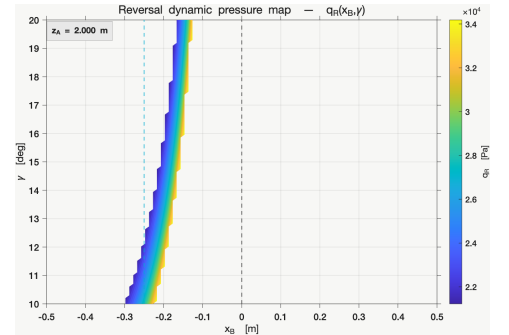
2) Along elastic axis ($x = 0$) : $z = (y/b)^2 \phi_w$, $\delta_z \approx 73\%$ – dominant bending reduction.

3) Chordwise variation of δ_z consistent with torsional term $-x(y/b)^2 \phi_\theta$ – strut simultaneously stiffens bending and torsion.


 Spanwise cuts of δ_z .

 Chordwise cuts of δ_z .

Effect of the strut on the reversal dynamic pressure

Physical reversal : first genuine zero of E_C with $0 < q_R < q_D$; sign changes near divergence singularity are not physical reversals.


 E_C vs. q , several x_B .

 q_R, q_D vs. x_B .

 Map $q_R(x_B, \gamma)$.

→ $x_B \geq 0$: E_C stays positive up to divergence – no physical reversal ; apparent sign changes near asymptotes are not genuine q_R .

→ $x_B < 0$: strut modifies bending–torsion coupling ; genuine q_R may appear.

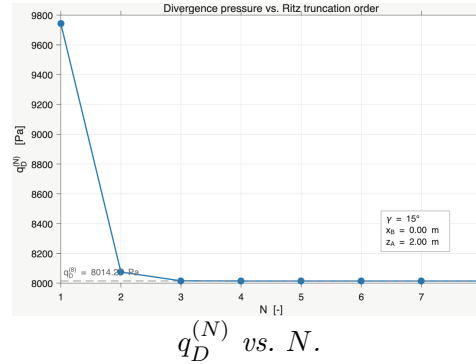
→ **Combined** q_R/q_D : beneficial configuration must raise both thresholds ; physical q_R exists only over a limited band of negative x_B .

→ **Map** : q_R confined near EA/AC for $x_B < 0$; increasing γ raises q_R – stronger coupling delays reversal.

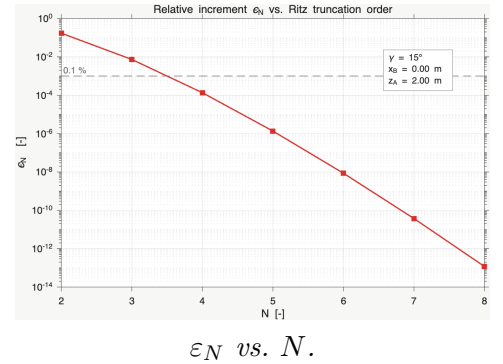
→ **Design objective** : maximize q_R or suppress reversal within flight envelope while keeping q_D high ; strut beneficial only for forward attachment with sufficiently large γ .

Convergence analysis of the Ritz approximation

$q_D^{(N)}$ drops sharply at low N then converges at $q_D \approx 8.014 \times 10^3$ Pa; no visible change beyond $N = 3$. ε_N falls by many orders of magnitude, reaching $\sim 10^{-14}$ at $N = 8$ – numerical saturation at machine precision, not additional physical accuracy.



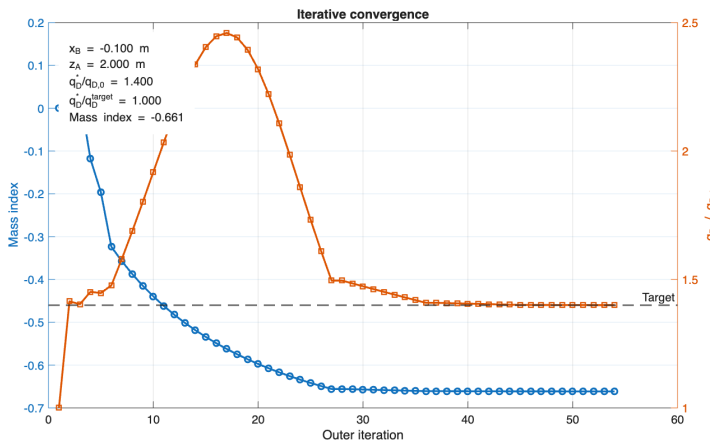
$q_D^{(N)}$ vs. N .



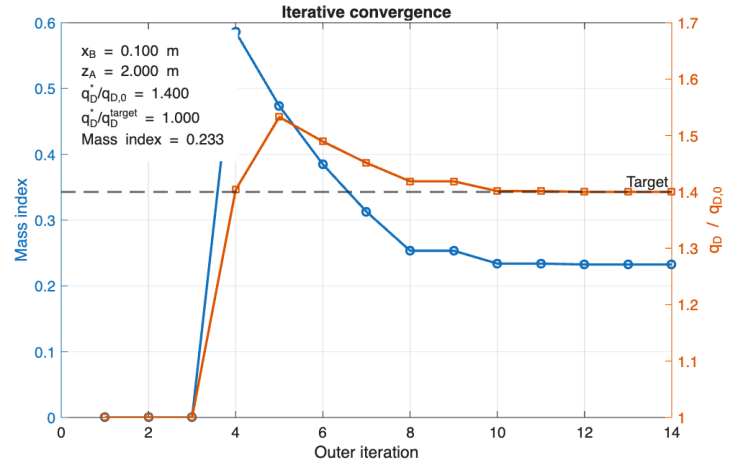
ε_N vs. N .

$N = 1$ overpredicts q_D by $\approx 22\%$: useful for trends but not quantitatively reliable. $N = 3-4$ already excellent; $N = 8$ merely confirms asymptotic convergence.

Iterative redesign results



Convergence, $x_B = -0.1$ m.



Convergence, $x_B = +0.1$ m.

	EJ/EJ_0	GJ/GJ_0	EA/EA_0	γ^*
$x_B = +0.1$	1.000	1.565	1.000	22.79°
$x_B = -0.1$	1.000	1.000	1.000	49.86°

	$q_{D,0}$ [Pa]	q_D^* [Pa]	$q_D^*/q_{D,0}$	$\mathcal{M}^*$
$x_B = +0.1$	6 687	9 363	1.400	+0.233
$x_B = -0.1$	18 348	25 688	1.400	-0.661

$\rightarrow x_B = +0.1$ m : target met by increasing GJ (+56%) and γ (+52%); EJ and EA unchanged. Mass cost $\mathcal{M} = +0.233$. The fact that EJ is not modified is physically consistent : divergence is governed by torsional stiffness, so bending stiffness has no predominant effect on q_D . EA is left unchanged as increasing it would raise the strut mass with limited benefit.

$\rightarrow x_B = -0.1$ m : only γ is modified ($15^\circ \rightarrow 49.86^\circ$, $\times 3.32$); all stiffness parameters unchanged; Net mass *reduction* ($\mathcal{M} = -0.661$) – the strut at higher γ provides stronger stabilization at lower structural cost and mass cost because it is shorter.

Forward attachment ($x_B < 0$) yields a far more efficient redesign – the target is achieved by geometry alone with a 66% mass reduction, versus added stiffness and +23% mass for $x_B > 0$.